

Shaoyuan Xie

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EDUCATION

University of California, Irvine

Ph.D. in Computer Science, Advisor: Professor Qi Alfred Chen

Irvine, CA

Expected Jun 2028

Huazhong University of Science and Technology (HUST)

B.Eng. in Automation, GPA: 3.97 (Top 1st / 212 students)

Wuhan, CN

Jun 2023

WORKING EXPERIENCE

Waymo

Research Intern, AI Foundation | Mentor: Chen Song, Hao Xiang, Yang Liu

- Develop and research Waymo Perception Foundation Model.

Mountain View, CA

Jun 2026 – Sep 2026

Mercedes-Benz Research & Development North America

Research Intern, AD Perception | Mentor: Thomas Monninger | Manager: Sihao Ding

- Semantic distillation from LLM to BEV encoder for safer end-to-end driving.

San Jose, CA

Jun 2025 – Sep 2025

Shanghai Artificial Intelligence Laboratory

Research Intern, OpenMMLab | Mentor: Wenwei Zhang | Manager: Kai Chen

- [InternLM](#) language models post training.

Shanghai, CN

Jun 2023 – Aug 2023

TOP-TIER PUBLICATIONS

Summary

2 (1 first-author) in top-tier machine learning conferences (NeurIPS, ICLR)

3 (3 first-author) in top-tier computer vision conferences and journals (TPAMI, ICCV, ECCV)

3 (1 first-author) in top-tier security conferences (NDSS)

1 in top-tier robotics conferences (IROS)

• BEVLM: Distilling Semantic Knowledge from LLMs into Bird's-Eye View Representations

Thomas Monninger*, Shaoyuan Xie* (*order determined alphabetically), Qi Alfred Chen, Sihao Ding
ECCV, 2026 (acceptance rate 27.5% = 2883/10473)

• FlyTrap: Physical Distance-Pulling Attack Towards Camera-based Autonomous Target Tracking Systems

Shaoyuan Xie, Mohamad Fakih, Junchi Lu, Fayzah Alshammari, Ningfei Wang, Takami Sato, Halima Bouzidi, Mohammad Al Faruque, Qi Alfred Chen
NDSS, 2026 (acceptance rate 17.9% = 265/1481)

• Are VLMs Ready for Autonomous Driving? An Empirical Study from Reliability, Data, and Metric Perspectives

Shaoyuan Xie, Lingdong Kong, Yuhao Dong, Chonghao Sima, Wenwei Zhang, Qi Alfred Chen, Ziwei Liu, Liang Pan
ICCV, 2025 (acceptance rate 24.0% = 2699/11239)

• Benchmarking and Improving Bird's Eye View Perception Robustness in Autonomous Driving

Shaoyuan Xie, Lingdong Kong, Wenwei Zhang, Jiawei Ren, Liang Pan, Kai Chen, Ziwei Liu
TPAMI, 2025 (Impact Factor = 20.4)

• Can We Trust Embodied Agents? Exploring Backdoor Attacks against Embodied LLM-Based Decision-Making Systems

Ruochen Jiao*, Shaoyuan Xie* (*co-first author), Justin Yue, Takami Sato, Lixu Wang, Yixuan Wang, Qi Alfred Chen, Qi Zhu
ICLR, 2025 (acceptance rate 32.1% = 3689/11500)

- **The Heat is On: Understanding and Mitigating Vulnerabilities of Thermal Image Perception in Autonomous Systems**
Hrushikesh Varma, Shaoyuan Xie, Michael Clifford, Qi Alfred Chen, Takeshi Sugawara, Sara Rampazzi
NDSS, 2026 (acceptance rate 17.9% = 265/1481)
- **Revisiting Physical-World Adversarial Attack on Traffic Sign Recognition: A Commercial Systems Perspective**
Ningfei Wang, Shaoyuan Xie, Takami Sato, Yunpeng Luo, Kaidi Xu, Qi Alfred Chen
NDSS, 2025 (acceptance rate 15.4% = 211/1369)
- **RoboDepth: Robust Out-of-Distribution Depth Estimation under Corruptions**
Lingdong Kong, Shaoyuan Xie, Hanjiang Hu, Lai Xing Ng, Benoit R. Cottureau, Wei Tsang Ooi
NeurIPS, 2023 (acceptance rate 26.1% = 3218/12343)
- **Adversarial Spatial Transform Marker: A Black-Box Pose Estimation Attack for Autonomous UAV Landing**
Junchi Lu, Fayzah Alshammari, Shaoyuan Xie, Xiaoqing Liang, Qi Alfred Chen
IROS, 2026

OTHER PEER-REVIEWED PUBLICATIONS

- **"Prompter Says": A Linguistic Approach to Understanding and Detecting Jailbreak Attacks Against Large-Language Models**
Dylan Lee, Shaoyuan Xie, Shagoto Rahman, Kenneth Pat, David Lee, Qi Alfred Chen
ACM CCS Workshop - LAMPS, 2024
- **On the Adversarial Robustness of Camera-based 3D Object Detection**
Shaoyuan Xie, Zichao Li, Zeyu Wang, Cihang Xie
TMLR, 2024

PRE-PRINT

- **SafeDriveVLA: Navigation-Conditioned World Model Dreaming for Conflict-Aware End-to-End Autonomous Driving**
Shaoyuan Xie*, Zihan Zhang*, Jingxuan Wang, Jiashu Qu, Xiaoqing Liang, Lingdong Kong, Junchi Lu, Henrik I Christensen, Qi Alfred Chen

RESEARCH IMPACT

- Email acknowledgements from DJI on autonomous tracking vulnerabilities
- Email acknowledgements from automotive manufacturer on traffic sign recognition vulnerabilities (One company among Tesla, Toyota, Nissan, Mazda, and Hyundai; no direct revelation of the company identity due to the anonymity agreement achieved during the responsible vulnerability disclosure process)

AWARDS AND HONORS

- Best Demo, 3rd USENIX Symposium on Vehicle Security and Privacy (2025, *selected for best demo rate 18.2% = 2/11*)
- Honours Bachelor's Degree (2023, *top 1% in univ.*)
- Outstanding Undergraduate Student (2021, *top 1% in univ.*)
- National Scholarship, Ministry of Education of China (2021, *top 0.2% nation-wide*)
- National Scholarship, Ministry of Education of China (2020, *top 0.2% nation-wide*)

ACADEMIC SERVICES

- *Competition Organizer*: IROS RoboSense (2025), ICRA RoboDrive (2024), ICRA RoboDepth (2023)
- *PC member*: ACM CCS Artifact Evaluation (2024, 2025, 2026), NDSS Artifact Evaluation (2025)
- *Reviewer*: CVPR (2023, 2024, 2025, 2026), ICCV (2025), ECCV (2026), BMVC (2026), ICLR (2026), CoRL (2026), IROS (2026), IJCV, RA-L, TDSC; TMM, TMLR
- *External Reviewer*: USENIX Security (2024, 2025), IEEE S&P (2024), ACM CCS (2024)